

Assignment 4

Due on May 25, 2026 (23:59:59)

Instructions: The main objective of this assignment is to build machine learning models that classify spoken words from silent lip movement videos.

Classification of Lip Movements on Videos

In this assignment, you will build machine learning models that classify spoken words from silent lip movement videos. Instead of training directly on raw video frames, which is computationally expensive, the assignment focuses on extracting facial landmark features from each frame, especially the mouth region. These landmarks will be converted into time-series representations of lip motion. To classify these lip movement sequences, you will experiment with different time-series classification approaches.

1. Dataset

In this assignment, you will use a public lipreading dataset for video-based word classification. The dataset consists of short video clips where each video contains a person pronouncing a sequence of words forming a short phrase. Unlike isolated word datasets, each video includes multiple words spoken continuously, which creates a more natural speaking scenario.

Each video is accompanied by an alignment file that specifies the exact time intervals for each spoken word. These alignment files indicate which word is being pronounced at a given time segment, allowing you to segment the video into word-level sequences if needed. Silent periods in the videos are explicitly labeled as “sil”, representing non-speaking intervals.

The dataset contains more than 50 different word classes, and the phrases vary across videos, meaning that the word order changes and does not follow a fixed pattern. This variability results in a non-homogeneous dataset, where class distributions and temporal structures may differ between samples. This makes the classification task more challenging and closer to real-world lipreading scenarios.

You can download the dataset from [this link](#).

1.1. Feature Extraction on Dataset

Instead of using raw video frames, you will extract facial landmarks with MediaPipe Face Mesh to reduce complexity while preserving important lip motion information.

MediaPipe Face Mesh detects facial landmarks for each frame. In this assignment, you will use only the landmarks around the mouth and lip region. Each selected landmark is represented by its x and y coordinates, turning every video into a time-series representation of lip movement. This approach lowers dimensionality, focuses on mouth motion rather than

full-frame appearance, and makes the data easier to use with sequence-based and time-series classification models.

Each processed video should be saved as a .npz file containing the extracted lip landmark coordinates, normalized values, frame indices, the word label, and the original video path.

2. Approach

In this assignment, you will explore different methods for classifying spoken words from lip movement sequences. Each video will first be converted into a multivariate time-series representation by extracting lip landmarks from each frame using MediaPipe Face Mesh. Each time step represents the mouth shape at a specific frame, while the full sequence represents the temporal movement pattern of the lips during speech.

- **Global Alignment Kernel (GAK) + SVM:** GAK is a similarity measure designed for time-series data. It compares two lip movement sequences by softly aligning them over time, even if the speaking speed or word duration differs between videos. The resulting kernel matrix is then used with an SVM classifier to separate the word classes.

Tutorials for GAK: <https://shota.io/2017/04/11/global-alignment-kernels.html>,

https://tslearn.readthedocs.io/en/stable/user_guide/kernel.html

- **Shapelets + MLP:** Shapelets are short and informative subsequences that represent important local lip movement patterns. The Shapelet Transform converts each lip landmark sequence into a fixed-length feature vector by measuring its distance to selected shapelets. An MLP classifier is then trained on these transformed features to predict the spoken word.

Tutorial for Shapelets: https://tslearn.readthedocs.io/en/latest/user_guide/shapelets.html

- **LSTM:** An LSTM network directly processes the sequence of lip landmarks over time. By learning how mouth shapes change from frame to frame, the LSTM can capture temporal dependencies in lip motion and classify the spoken word.
- **Suggested Method:** You will also propose and implement one additional classification approach suitable for time-series data. For example, you may use a feature-based method that extracts statistical and temporal descriptors from lip landmark sequences, or another sequence classification model learned in class.

3. What should you do?

- Download the dataset from the provided Kaggle link.
- Extract lip landmark features from all videos using MediaPipe Face Mesh. Focus only on the mouth and lip region. Convert each video into a time-series representation of lip movements and save the extracted features as .npz files.

- Prepare the data by:
 - Loading the lip landmark sequences,
 - Applying normalization to reduce variations in scale and position,
 - Padding or truncating sequences to a fixed length,
 - Encoding the labels and splitting the dataset into training and test sets.
- Use the provided alignment files to segment videos into word-level sequences. You may ignore segments labeled as “sil” (silent).
- Implement the classification approaches using the extracted lip movement features:
 - Global Alignment Kernel (GAK) combined with an SVM classifier
 - Shapelet Transform followed by an MLP (PyTorch for MLP)
 - LSTM-based sequence classification model (PyTorch for LSTM)
 - Propose and implement one extra approach that uses a classification method covered in the course and is suitable for time-series data.
- Train and evaluate each model on the word classes. Perform a detailed ablation study for every approach. Experiment with different parameters and settings (such as kernel parameters, number and length of shapelets, MLP architecture, LSTM layers, or hidden sizes). Analyze how these changes affect performance instead of reporting only a single experiment per method.
- Present your results clearly, including:
 - Accuracy scores,
 - Confusion matrices,
 - Tables or plots comparing different configurations.
- Compare all methods and discuss their performance, highlighting strengths, limitations, and common failure cases. In your report, evaluate the methods based on factors such as ease of implementation, classification performance, computational efficiency, and their suitability for lip movement-based time-series data.

Submit

You are required to submit all your code in a Jupyter notebook, along with a report in ipynb format, which should also be prepared using Jupyter notebook. The code you submit should be thoroughly commented. Your report should be self-contained and include a concise overview of the problem and the details of your implemented solution. Feel free to include pseudocode or figures to highlight or clarify specific aspects of your solution. Finally, prepare a ZIP file named `project_studentIDs.zip` containing:

- Please write the your name and surname on your Jupyter notebook.

- project.ipynb (including your report and code)
- project.py (py file version of your ipynb file)
- **Do not send the dataset.**

The ZIP file will be submitted via [Submit](#).

Grading

- Code (50): Feature extraction (5 points), GAK+SVM (10 points), Shapelets+MLP (10 points), LSTM (15 points), Suggested method (10 points)
- Report(50): GAK+SVM experiments (10 points), Shapelets+MLP experiments (10 points), LSTM experiments (10 points), Suggested method experiments (10 points), Experiment Results Comparison (10 point)

Notes for the report: Preparing a good report is important as well as your solutions!

Academic Integrity

All work on assignments must be done individually unless stated otherwise. You are encouraged to discuss with your classmates about the given assignments, but these discussions should be carried out in an abstract way. That is, discussions related to a particular solution to a specific problem (either in actual code or in the pseudocode) will not be tolerated. In short, turning in someone else's work, in whole or in part, as your own will be considered as a violation of academic integrity. Please note that the former condition also holds for the material found on the web as everything on the web has been written by someone else.